

UNNATI YADAV

AI/ML Engineering Student · Generative AI · NLP · Agentic Systems

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SUMMARY

B.Tech CSE student (2023–2027) specialising in Generative AI and NLP, with hands-on experience building end-to-end agentic systems, voice AI pipelines, and diffusion-based image generation tools. DeepLearning.AI certified practitioner consistently achieving 95%+ in professional certifications. Published researcher in medical image classification.

RESEARCH / PUBLICATIONS

Skin Disease Classification Using Deep Learning · Under Review · 2025

- Developed a CNN-based multi-class classifier for skin disease detection; applied transfer learning and image augmentation techniques to improve generalisation on dermatological datasets.

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript

ML / DL: TensorFlow, PyTorch, Scikit-learn, Keras

Generative AI: LLMs (GPT-4o, Mistral-7B), Hugging Face Transformers, Stable Diffusion, ControlNet, Prompt Engineering

NLP & Speech: Resume Parsing, Sentiment Analysis, Speech Recognition (Whisper), NLP Pipelines

Tools & Infra: Streamlit, FastAPI, Git, GitHub, Jupyter, VS Code, Pydantic, Agentic AI

Data Science: Pandas, NumPy, Matplotlib

Math Foundations: Probability, Statistics, Linear Algebra, Calculus

PROJECTS

HR Resume & LinkedIn Shortlisting Agent | Python · LLMs (GPT-4o) · Streamlit · NLP · Pydantic

- Engineered an AI HR agent ranking candidates via a weighted 5-dimension scoring framework, cutting manual screening effort by ~70%.
- Built NLP parsing pipelines for resumes and LinkedIn profiles, extracting skills, education, and experience with >90% field accuracy.
- Implemented LLM evaluation workflow with structured JSON outputs, explainable scoring, override logging, and automated HTML/JSON reports.
- Designed Streamlit recruiter dashboard enabling real-time candidate comparison across 10+ attributes.
- GitHub: [https://github.com/Unnatiy200/hr_shortlist_agentA]

AI Recruiter Voice Agent | Python · TensorFlow · NLP · Whisper Speech Recognition

- Built end-to-end AI interview system dynamically generating personalised questions via resume parsing, reducing interviewer prep time by ~60%.
- Integrated real-time speech recognition + sentiment analysis pipeline evaluating 5 communication dimensions per candidate response.
- Fine-tuned offline Whisper-based speech model on open-source datasets, achieving 85% transcription accuracy without external API dependency.
- GitHub: [https://github.com/Unnatiy200/RECRUITER_VOICE_AGENT]

Interior AI Design Generator | Python · Stable Diffusion · ControlNet · GenAI

- Developed GenAI system producing photorealistic room renders from natural language prompts, cutting design ideation time by ~80%.
- Integrated ControlNet conditioning to preserve spatial geometry; built prompt pipeline mapping style, mood, and colour to optimised diffusion inputs.
- Extended expertise from skin disease classification research (CNNs, transfer learning) to diffusion-based image generation pipelines.
- GitHub: [https://github.com/Unnatiy200/interior_ai_design_generator]

EDUCATION

B.Tech in Computer Science & Engineering — Bennett University, Uttar Pradesh Aug 2023 – Aug 2027

CGPA: 7.7 / 10 ($\approx$ 3.5 / 4.0)

Relevant Coursework: Machine Learning, Deep Learning, Probability & Statistics, Data Structures & Algorithms, Linear Algebra

CERTIFICATIONS

Neural Networks and Deep Learning — DeepLearning.AI · Coursera

Apr 2025

Grade: 98.60% · Topics: Forward/back propagation, activation functions, gradient descent, deep network optimisation

Apply GANs: Pix2Pix & CycleGAN — DeepLearning.AI · Coursera

Apr 2026

Topics: Conditional image generation, domain transfer, discriminator training, evaluation metrics (FID, IS)

Next-Gen Workplace Automation with GEN-AI (Specialisation — 8 courses) — Coursera Apr 2026